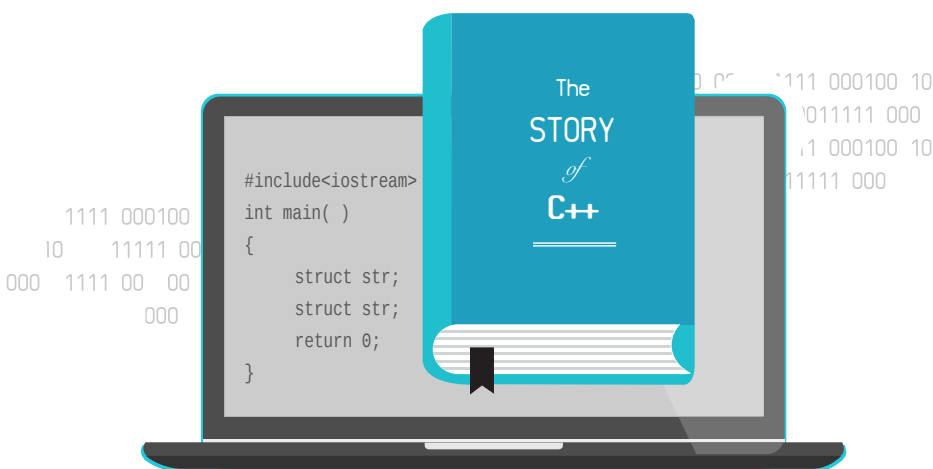


Different C++ Standards: The Story of C++

C++ is a programming language very popular in academia and the industry. When a professor tries to cover a lot of topics within a short span of three to four months, it is natural to omit the history part of a programming language altogether. This may not be a good strategy in the long run, because to fully understand the intricacies and power of a programming language some knowledge of its history is absolutely essential. This article addresses the historic aspects of C++ and the different standards over the years. It serves as good supplementary reading material to students learning C++ for the first time.



First, let's begin with the origin of C++ and also about the man behind it, Bjarne Stroustrup. C++ is one of the most popular programming languages in the world, and has been a permanent fixture in the top ten of every programming language popularity ranking scheme for many years. It is interesting to know that all the top four programming languages in the April 2021 ranking of the TIOBE Index were developed by individuals. They are: Dennis Ritchie (C), Guido van Rossum (Python), James Gosling (Java) and Bjarne Stroustrup (C++). All of them are superstars in the world of computer programming.

Nowadays programming languages are designed by a team of computer scientists and the credit goes to the parent organisation. But more than 30 years ago, the task was done by lone geniuses. Bjarne Stroustrup is a Danish computer scientist. While working at AT&T Bell Laboratories, he had many bright ideas to improve the ever popular C programming language. By the end of the 70s (a time when C was just celebrating its ninth birthday), Stroustrup presented his improvements to C. At first he considered this new language just as an extension to C and was modest enough to call it 'C with Classes'. Soon, the new language started gaining the attention of serious

programmers and by the early 80s people started considering it as a programming language independent of C.

The name C++ comes from the programming language C and the post-increment operator '++' used in C as well as C++. The procedural programming features of C++ were influenced by C, and the object-oriented programming features of C++ were influenced by a programming language called Simula. For almost two decades, the *de facto* standard of C++ was set by the textbooks on C++ written by Bjarne Stroustrup. Finally, in 1998 came the first standard of C++ called C++98. C++11 and C++17 standards brought in several major changes and additions to the C++ language. In between C++03 and C++14 brought in minor changes. C++20 is the latest standard of C++, and as of now no compiler has full support for all the features mandated by this standard. The next standard of C++ is the proposed C++23. Though begun as an extension to C, C++ today is an object-oriented programming language with additional support for some functional programming features.

Now, let us discuss the different standards of C++ in detail. To enhance our understanding about each of these standards, we will also discuss small